LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Tyre and brake care
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Tyre and brake care
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Inspection of aircraft tyre and brake assembly; Safety: Wear eye protection and gloves; Supervision: Instructor monitors pressure gauge readings and torque application.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft landing gear?
(b) Name two common types of aircraft tyres.
(c) Why is regular brake maintenance critical for flight safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Main landing gear structural components • Tyre construction layers and materials • Brake assembly types and mechanisms • Tyre pressure monitoring systems • Brake pad wear indicators	L & Ds	PPT, IB	15 min
(b)	• Tyre inflation procedures and limits • Brake system inspection checklist • Identification of common tyre defects • Brake fluid contamination causes • Wheel bearing lubrication standards	L & Ds	PPT, PDF	20 min
(c)	• Post-flight tyre and brake checks • Reporting defects to maintenance logs • Tyre rotation and replacement criteria • Brake overheating prevention techniques • Storage requirements for spare tyres	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a bias-ply and a radial tyre construction.
(b) Describe the steps to identify a cracked brake pad during inspection.
(c) How does incorrect tyre pressure affect aircraft braking performance?

12. Home Assignment and Follow Up	Prepare a brief report on the basic design requirements and advantages of a typical pneumatic system to be discussed in Lesson Plan 19.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min